



Home Assistant

Community Survey 2024

Protocol Usage -
Zigbee

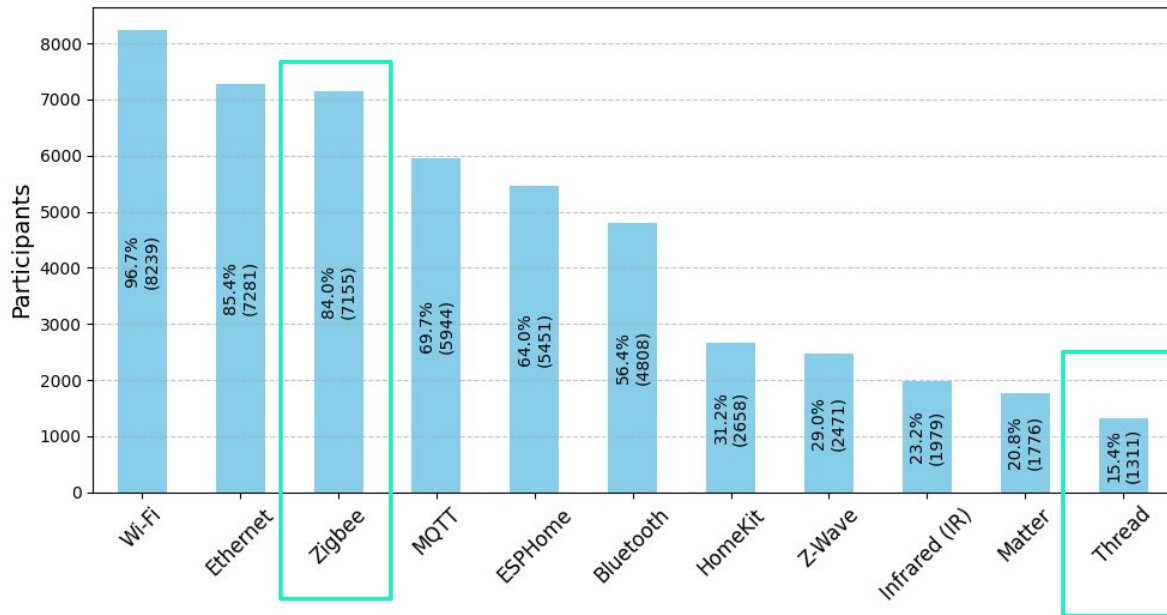
03/26/25
Annika Schulz

Most used protocols (multiple choice)

- Zigbee is the third most used protocol of the community survey participants
- Thread is not part of the top 10

Top 11 Used Communication Protocols

8522 out of 8619 people answered this question

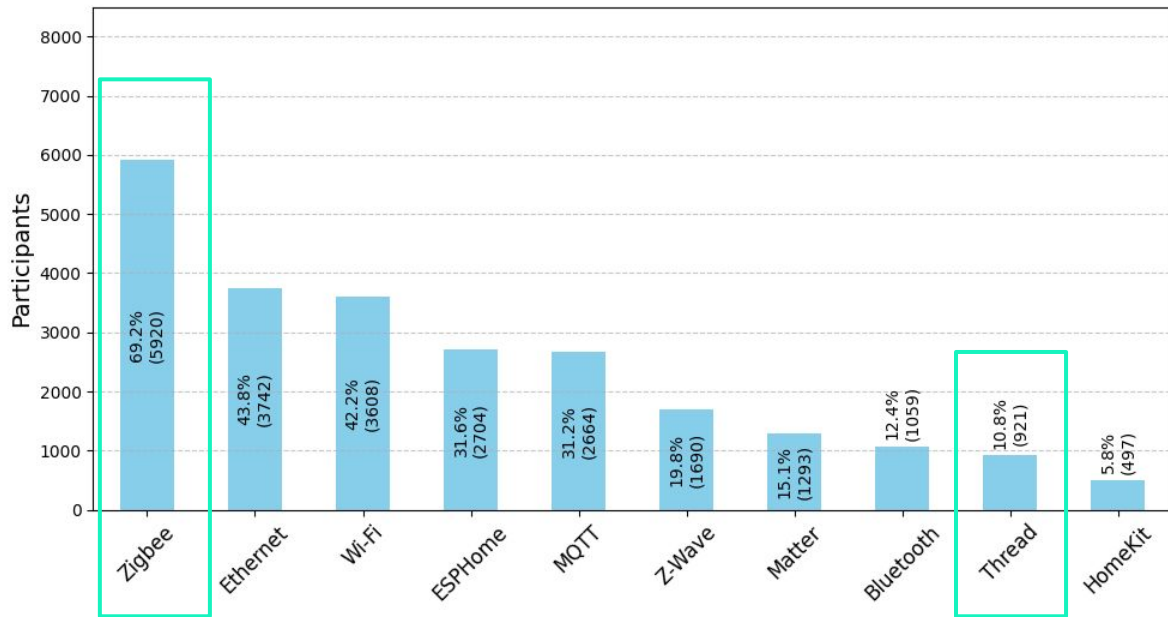


Favorite protocols (multiple choice)

- Zigbee is the favorite protocol of the participants
- Thread is number 9

Top 10 Favorite Communication Protocols

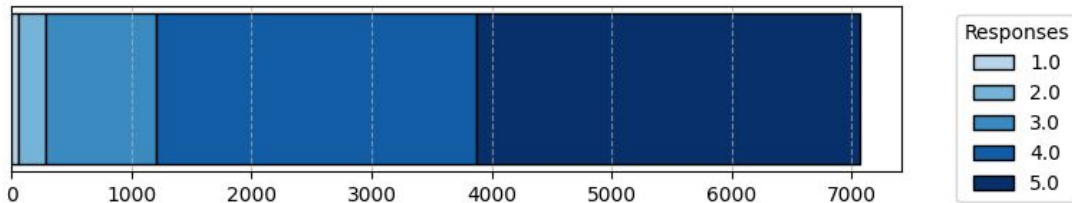
8552 out of 8619 people answered this question



Reliability

- Users perceive the zigbee network as reliable
- Thread is not perceived as as reliable in comparison

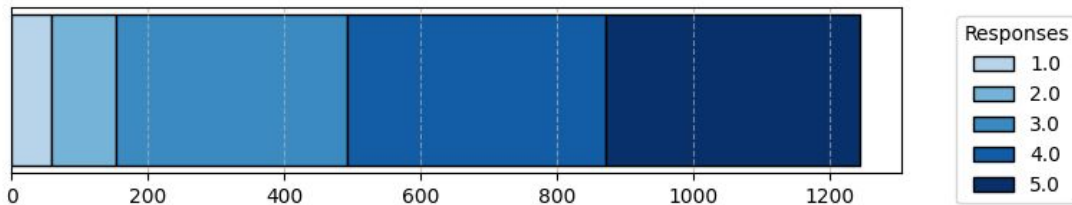
The Zigbee network at my home is reliable.



7068 out of 8619 people answered this question

Mean Likert Score: 4.23
Standard Deviation: 0.85
Median: 4.00

The Thread network at my home is reliable.



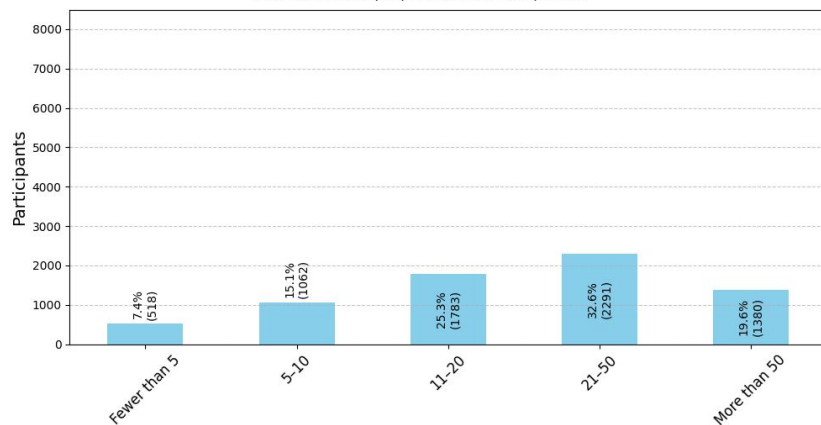
1245 out of 8619 people answered this question

Mean Likert Score: 3.73
Standard Deviation: 1.11
Median: 4.00

of Zigbee and Thread devices

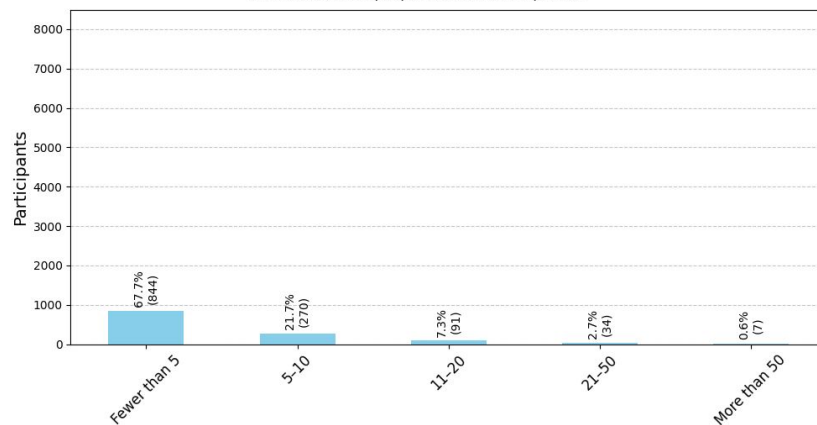
Approximately how many Zigbee or Thread devices do you have in your home?

7034 out of 8619 people answered this question



Approximately how many Thread devices do you have in your home?

1246 out of 8619 people answered this question

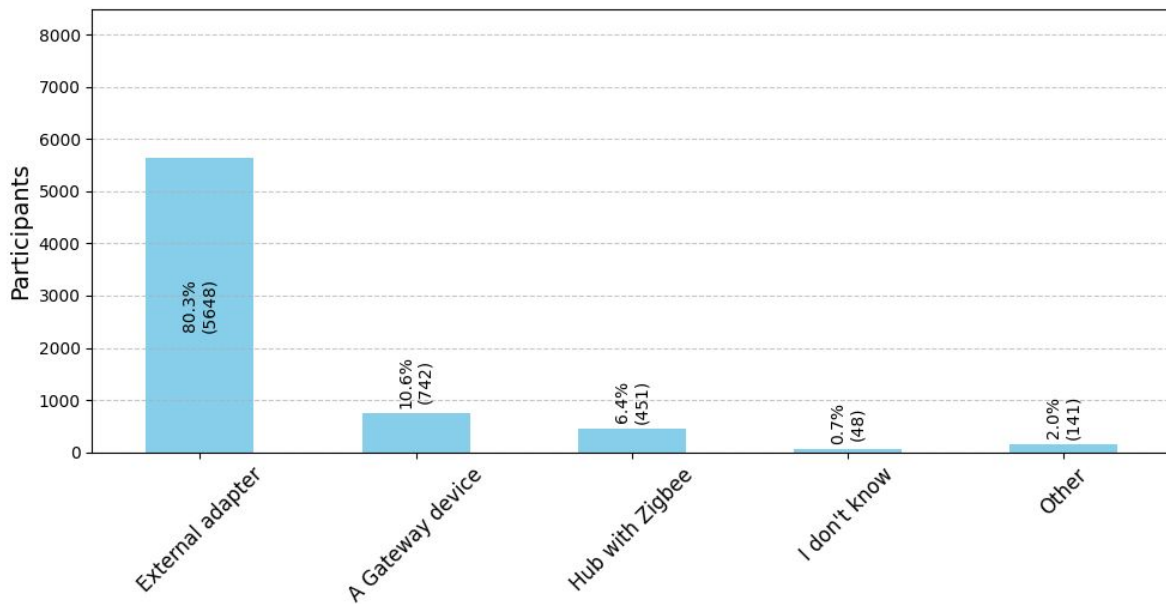


Hubs used for Zigbee Network

- Several people use multiple hubs
- Many started with Philips Hue or IKEA and stick to it

What kind of hub are you using for your Zigbee network?

7030 out of 8619 people answered this question



"I run four separate Zigbee networks — three using external Zigbee adapters via Zigbee2MQTT, and one through a Hue bridge." (*paraphrased*)

Hue hub, Dirigera (IKEA) hub, External adapter

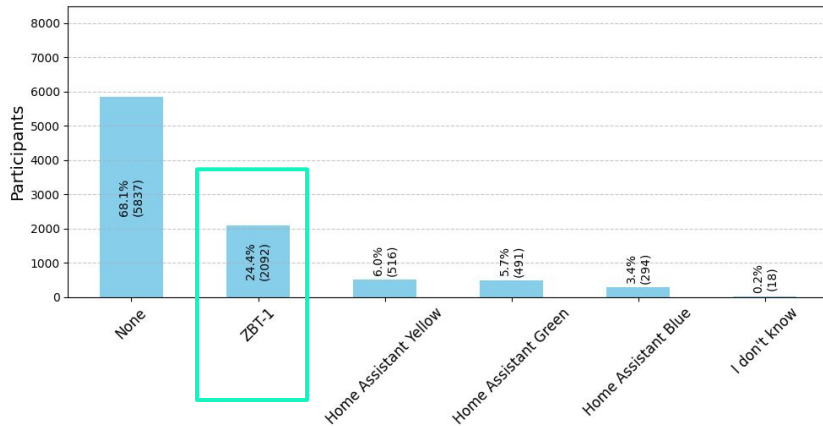


Home Assistant

More ZBT-1s were purchased than used.

Purchased Nabu Casa Hardware

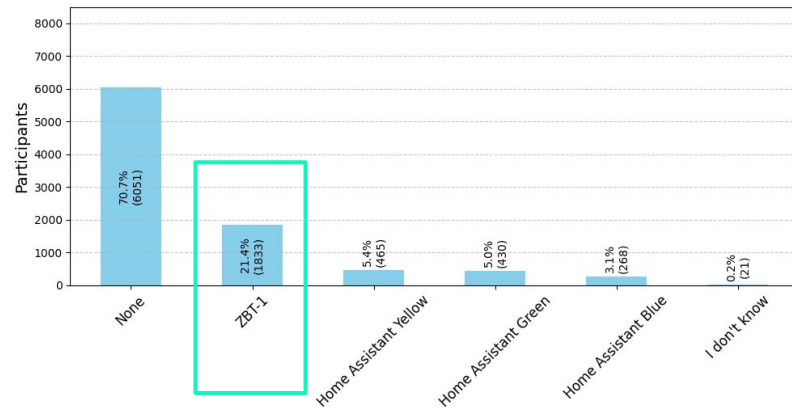
8566 out of 8619 people answered this question



"I've got a Green and a SkyConnect on order, and plan to move to ZHA once they arrive." (paraphrased)

Currently Using or Have Used Nabu Casa Hardware

8554 out of 8619 people answered this question



Most used Zigbee Software Stack

- Zigbee2MQTT is more often used, although ZHA is the default
- Several people use multiple software stacks
- Assumption: People using hubs with integrated zigbee stacks don't know much about the software stack.

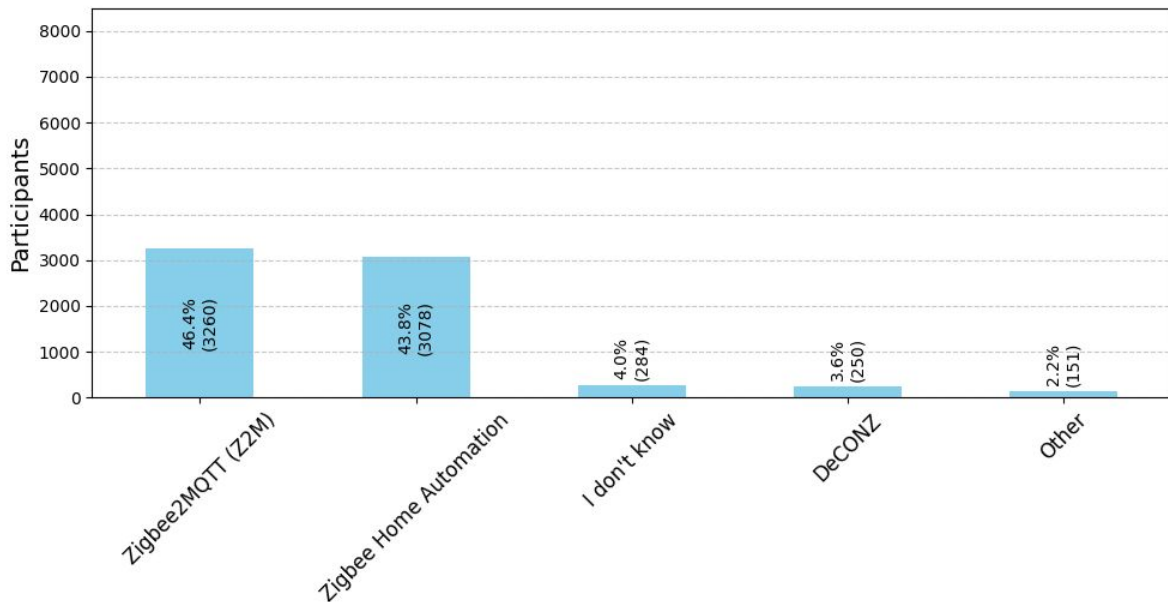
Hue was the most common answer for other SW stacks

"ZHA + transitioning to Z2M"
(paraphrased)

"Z2M, transitioning to ZHA"
(paraphrased)

What Zigbee software stack are you currently using?

7023 out of 8619 people answered this question



Reasons for software stacks

- Often, people start with the default protocol
- Many people stick to their earlier decisions if they work well enough
- Some users run multiple networks using different stacks - migration seems to involve a lot of effort

If they consciously choose a protocol or migrate, it's often Zigbee2MQTT for various reasons:

Customizability and more features

Decoupling from Home Assistant for improved stability

Device support / device database and documentation

Recommendations

GUI and network visualization

Ability to expose more entities

Reliability and stability

MQTT integration



Aspects of Zigbee network to improve (I)



Stability & Performance

Reliability: Overall reliability and stability of their Zigbee networks, with fewer device disconnections and dropouts.

Performance: Faster response times and lower latency, especially for lighting control.

Interference: Reducing interference with WiFi and other wireless networks.



Coverage & Network Management

Range Expansion: Expanding network coverage, especially to outdoor areas, basements, and detached buildings is a common desire.

Mesh Optimization: Better control over routing and mesh topology.

Network Visualization: Improved network mapping and visualization tools.



Aspects of Zigbee network to improve (II)

Device Support & Compatibility

Device Compatibility: Better support for more devices and features for ZHA compared to Zigbee2MQTT.

Binding: Easier setup of device-to-device bindings.

Firmware Updates: Easier and more reliable firmware updates for devices.

Battery Life: Improving battery life of sensors and other battery-powered devices.

Usability & Troubleshooting

Troubleshooting Tools: Better tools and visibility for diagnosing network issues and problematic devices.

Backup: Improved backup and restore capabilities, especially when changing coordinators.

Migration: Simpler migration between different Zigbee software stacks (e.g. ZHA to Z2M).

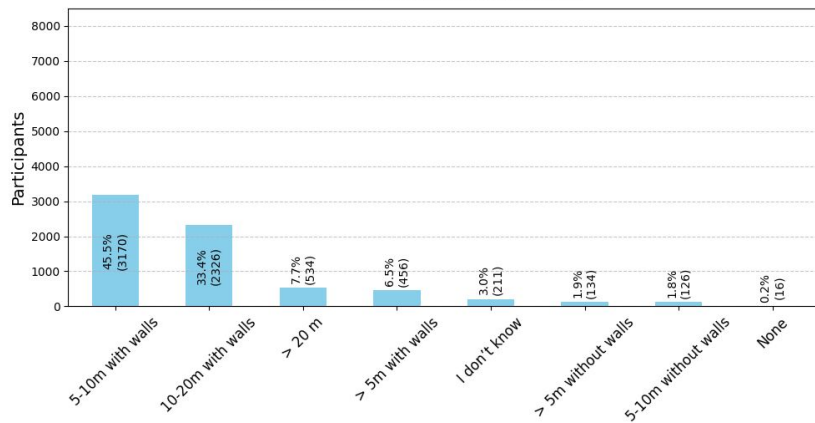
Notifications: Alerts when devices go offline or have low batteries.



Zigbee Device Distance from Hub

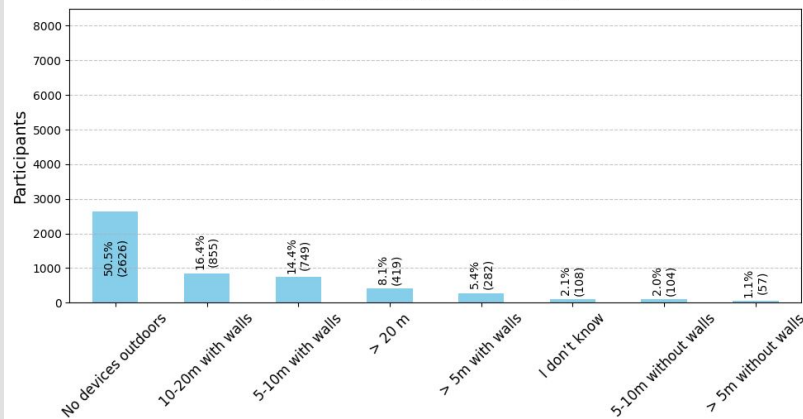
How far is the furthest Zigbee device from your hub? - Indoors

6973 out of 8619 people answered this question



How far is the furthest Zigbee device from your hub? - Outdoors

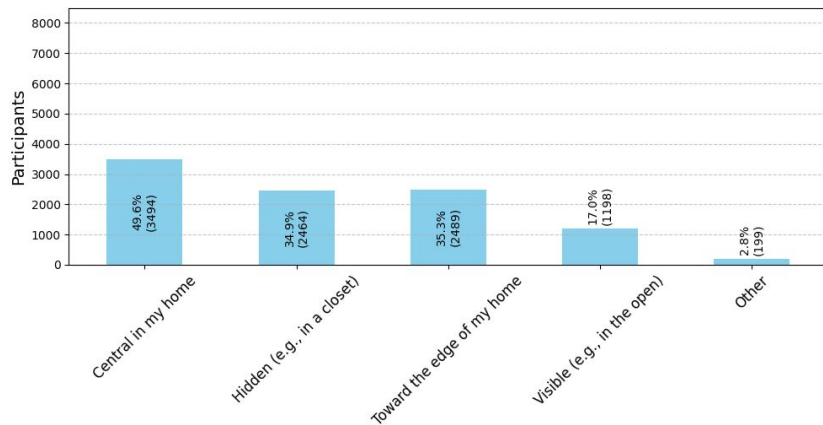
5200 out of 8619 people answered this question



Zigbee Hub Location

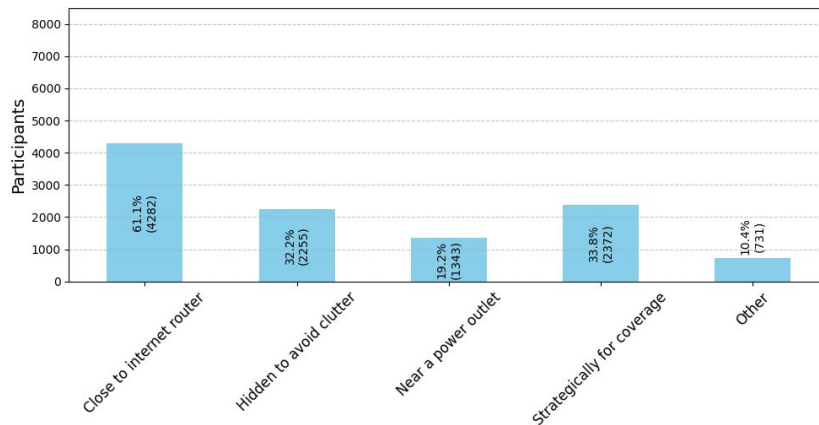
Where is your Zigbee hub located?

7051 out of 8619 people answered this question



Why did you choose this location for your Zigbee hub?

7012 out of 8619 people answered this question



Zigbee Hub Locations - "Other"

Basement

Server room

Utility room

Pantry

Clothing closet

Outside the home

Attic

Roof space

Loft

Workshop

Garage

Office

Outside the home

In a cabinet

In a server rack

behind the TV

on top of sth.
e.g. cabinet, rafters

below sth.,
e.g., desk, staircase

at the ceiling

in an alcove

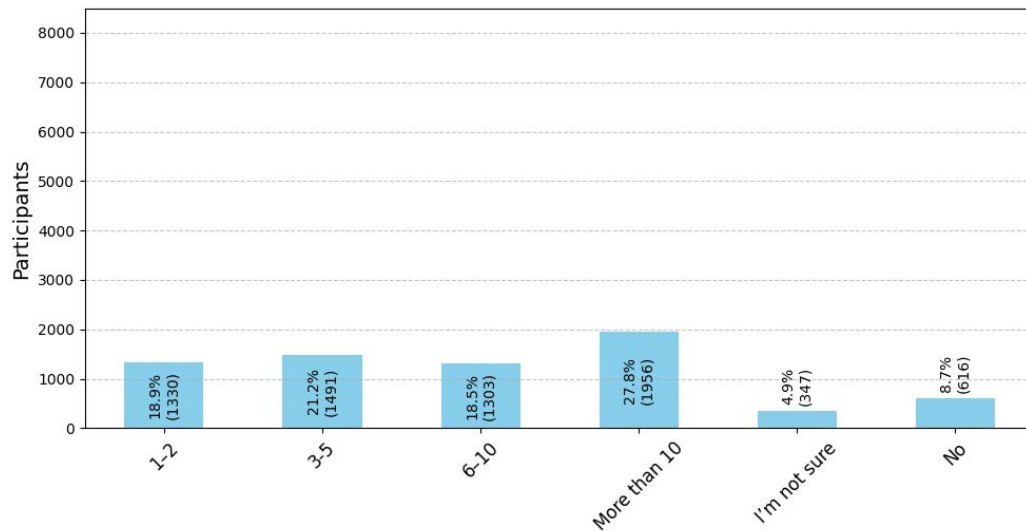


Zigbee Routers

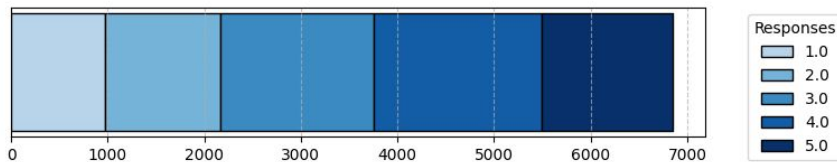
- Most people have numerous Zigbee routers, most common more than 10
- Some people place them strategically, but it's not common - the Likert scale question was rated neutral

How many Zigbee routers (e.g., repeaters or routing-capable devices) do you use in your network?

7043 out of 8619 people answered this question



I strategically consider the placement of Zigbee routers to optimize the mesh network.



6846 out of 8619 people answered this question

Mean Likert Score: 3.19
Standard Deviation: 1.32
Median: 3.00